

Question Bank
NATURAL LANGUAGE PROCESSING
(CML23701)

Complied By:

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Module 1

Q. No.	Questions	CO's	RBT Level	Marks
Unit 1: Overview				
1.	Describe the motivation behind the development of computational models of languages.	CO1	L2	5 M
2.	Describe and compare the rationalist and empiricist approaches used in NLP, highlighting their major differences.	CO1	L2	10 M
3.	Write a note on origins of NLP.	CO1	L2	5 M
4.	Explain the different levels of linguistic analysis in NLP with suitable examples.	CO1	L2	10M
5.	Write a note on role of knowledge in language understanding.	CO1	L2	5 M
6.	Explain the challenges that makes natural language processing difficult.	CO1	L2	10 M
7.	Describe different types of grammars in NLP, with examples.	CO1	L2	10 M
8.	Describe the two dimensions of sentence structure in NLP and explain their basic components with suitable examples	CO1	L2	10 M
9.	Describe the components of transformational grammar.	CO1	L2	5 M
10.	Describe the three main components of Transformational Grammar and explain the function of each component with suitable examples.	CO1	L2	10 M
11.	Describe some successful early NLP systems with their key features and applications.	CO1	L2	10 M
12.	What is Generative Grammar? Explain its basic concepts and components with suitable examples	CO1	L2	10 M
Unit 2: Language Modeling				
13.	Explain the importance of language modeling in NLP.	CO1	L2	5 M
14.	Explain the major challenges faced in developing NLP systems for Indian languages, with suitable examples.	CO1	L2	10 M
15.	Describe the major applications of NLP and explain how they are used in real-world scenarios with suitable examples	CO1	L2	10 M
16.	What is Information Retrieval? Explain its basic process, key characteristics, and major issues.	CO1	L2	10 M
17.	Explain the Theta-Criterion and Theta Grid in GB Theory with suitable examples.	CO1	L2	10 M
18.	What is Hierarchical Grammar? Explain its basic structure and key features with suitable examples.	CO1	L2	10 M
19.	Explain Phonetic Form (PF) and Logical Form (LF) in Government and Binding (GB) Theory, along with its key principles	CO1	L2	10 M
20.	Explain X-Bar Theory in Government and Binding (GB) Theory, including its basic principles and components with suitable examples.	CO1	L2	10 M
21.	List and explain common thematic roles (Θ-Roles)	CO1	L2	5 M
22.	What are Case Theory and the Case Filter? Explain the different types of Case with suitable examples.	CO1	L2	10 M

23.	Briefly explain the Lexical-Functional Grammar (LFG) model and its key components with suitable examples.	CO1	L2	10 M
24.	Explain the Paninian Framework in NLP with its key concepts and suitable examples.	CO1	L2	10 M
25.	Describe the different types of Statistical Language Models with suitable examples.	CO1	L2	10 M

Module 2

Q. No.	Questions	CO's	RBT Level	Marks
Unit 1: Word Level Analysis				
1.	What are Regular Expressions? Explain their basic building blocks with suitable examples.	CO2	L2	10 M
2.	Explain character classes in regular expressions with suitable examples	CO2	L2	5 M
3.	Write a regular expression to check a string is an email address or not.	CO2	L3	10 M
4.	Describe the different types of Finite Automata with suitable examples.	CO2	L2	10 M
5.	Explain the two broad classes of morphemes with suitable examples.	CO2	L2	10 M
6.	Explain the three main ways of word formation—Inflection, Derivation, and Compounding—with suitable examples.	CO2	L2	10 M
7.	Explain a Morphological Parser and describe its major information sources with suitable examples.	CO2	L2	10 M
8.	Explain the major limitations of Morphological Parsers with suitable examples	CO2	L2	10 M
9.	Explain the Two-Level Morphological Model, including its structure, working, and key features with suitable examples.	CO2	L2	10 M
10.	Draw a simple finite-state transducer that maps the input strings hot and cat to cot and bat, respectively.	CO2	L3	5 M
11.	Explain the major spelling correction algorithms with suitable examples	CO2	L2	10 M
12.	Explain the two approaches to spelling error detection and correction with suitable examples.	CO2	L2	10 M
13.	Write the Minimum Edit Distance algorithm and apply it to find the edit distance between 'cat' and 'cut.	CO2	L3	10 M
Unit 2: Syntactic Analysis				
14.	Explain the different word classes with suitable examples.	CO2	L2	5 M
15.	Explain why Part-of-Speech (POS) Tagging is challenging, with suitable examples.	CO2	L2	5 M
16.	Explain the rule-based approach to Part-of-Speech (POS) tagging with a suitable example.	CO2	L2	10 M
17.	Illustrate N-gram taggers for POS tagging with suitable examples.	CO2	L3	10 M
18.	Illustrate the Hidden Markov Model (HMM) and its application in POS tagging with suitable examples.	CO2	L3	10 M
19.	Explain Context-Free Grammar (CFG) with sentence derivation and parse tree using a suitable example.	CO2	L2	10 M
20.	Explain phrase-level constructions in NLP with suitable examples	CO2	L2	10 M
21.	Apply the shift-reduce parsing algorithm to parse the sentence 'The cat chased a dog' using the given grammar. 1. $S \rightarrow NP VP$ 2. $NP \rightarrow Det N$ 3. $VP \rightarrow V NP$ 4. $Det \rightarrow 'The' \mid 'a'$	CO2	L3	10 M

	5. $N \rightarrow \text{'cat'} \mid \text{'dog'}$ 6. $V \rightarrow \text{'chased'} \mid \text{'saw'}$			
22.	Apply top-down depth-first parsing to parse the sentence 'The boy eats an apple' using the following grammar. Show the parsing steps. 1. $S \rightarrow NP VP$ 2. $NP \rightarrow Det N$ 3. $VP \rightarrow V NP$ 4. $Det \rightarrow The \mid an$ 5. $N \rightarrow boy \mid apple$ 6. $V \rightarrow eats$	CO2	L3	10 M
23.	Apply the Earley parsing algorithm to parse the sentence 'A girl reads the book' using the following grammar. Show the parsing steps. 1. $S \rightarrow NP VP$ 2. $NP \rightarrow Det N$ 3. $VP \rightarrow V NP$ 4. $Det \rightarrow a \mid the$ 5. $N \rightarrow girl \mid book$ 6. $V \rightarrow reads$	CO2	L3	10 M
24.	Apply the CYK parsing algorithm to parse the sentence 'The girl wrote an essay' using the following grammar. Show the parsing steps. 1. $S \rightarrow NP VP$ 2. $VP \rightarrow Verb NP$ 3. $NP \rightarrow Det Noun$ 4. $Det \rightarrow an \mid the$ 5. $Verb \rightarrow wrote$ 6. $Noun \rightarrow girl$ 7. $Noun \rightarrow essay$	CO2	L3	10 M
25.	Discuss the applications of syntactic analysis in NLP tasks.	CO2	L2	10 M

Module 3

Q.No.	Questions	CO's	RBT Level	Marks
Unit 1: Semantic Analysis				
1.	Explain the importance of semantic analysis in NLP.	CO3	L2	5 M
2.	Explain the major challenges in Semantic Analysis in NLP with suitable examples.	CO3	L2	10 M
3.	Explain the characteristics of a good meaning representation with suitable examples.	CO3	L2	10 M
4.	Describe the fundamental predicate–argument structure with suitable examples.	CO3	L2	10 M
5.	Explain syntax-driven semantic analysis with a neat diagram and suitable example.	CO3	L2	10 M
6.	Explain lambda expressions and lambda reduction in semantic analysis with suitable examples.	CO3	L2	10 M
7.	Illustrate the semantic analysis approaches for the sentence ‘President nominates speaker’ with suitable examples.	CO3	L3	10 M
8.	Write a note on semantic grammars and explain their use with suitable examples.	CO3	L2	5 M
9.	What is Lexical Semantics? Explain the relationships among lexemes and their senses with examples.	CO3	L2	10 M
10.	Explain the internal structure of words with suitable examples.	CO3	L2	10 M
11.	What is theta role? Describe some commonly recognized thematic roles.	CO3	L2	10 M
12.	Write the WordNet senses of the word ‘apple’ with suitable examples.	CO3	L3	5 M
13.	What is ambiguity? Explain the four main levels of ambiguity with suitable examples.	CO3	L2	10 M
Unit 2: Natural Language Generation				
14.	Describe natural language generation with example and list some applications.	CO3	L2	10 M
15.	Illustrate Pipelined NLG architecture with example and list advantages and disadvantages.	CO3	L3	10 M
16.	Illustrate Interleaved NLG architecture with example and list advantages and disadvantages.	CO3	L3	10 M
17.	Analyze the given sentence step by step using discourse planning principles: “Savitha sang a song. The song was good although some people do not like it.”	CO3	L3	10 M
18.	With neat diagram briefly explain input and output of three NLG tasks.	CO3	L2	10 M
19.	Write a note on the role of the micro planner.	CO3	L2	05 M
20.	Explain sentence aggregation in a micro-planner with suitable examples.	CO3	L2	10 M
21.	Explain lexicalization in a micro-planner with suitable examples.	CO3	L2	10 M

22.	Describe referring expression generation with suitable examples.	CO3	L2	10 M
23.	What is surface realization? Explain the role of surface realization with example.	CO3	L2	10 M
24.	Illustrate the layers of Systemic Functional Grammar (SFG) for the sentence 'savitha sang a song'.	CO3	L3	10 M
25.	Explain Functional Unification Grammar (FUG) for surface realization with example.	CO3	L2	10 M
26.	Describe applications of Natural Language Generation (NLG) with examples.	CO3	L2	10 M

Module 4

Q.No.	Questions	CO's	RBT Level	Marks				
Unit 1: Machine Translation								
1.	What is machine translation give examples and list its applications.	CO4	L2	5 M				
2.	Describe problems in machine translation.	CO4	L2	10 M				
3.	Summarize the characteristics of Indian languages.	CO4	L2	10 M				
4.	Illustrate with neat diagram how direct machine translation approach works.	CO4	L3	10 M				
5.	Explain the main components of rule-based machine translation approach with example.	CO4	L2	10 M				
6.	Describe how transfer-based machine translation works.	CO4	L2	10 M				
7.	Illustrate how interlingua-based machine translation works with neat figure.	CO4	L3	10 M				
8.	Summarize the advantages and disadvantages of interlingua-based machine translation.	CO4	L2	5 M				
9.	Describe types of Rule-Based machine translation systems.	CO4	L2	5 M				
10.	Translate the input string “Khushbu slept in the garden” using interlingua-based machine translation.	CO4	L3	5 M				
11.	Explain the key components of an Information Retrieval system.	CO4	L2	10 M				
Unit 2: Information Retrieval								
12.	Illustrate the entities involved in an Information Retrieval system with suitable examples.	CO4	L3	10 M				
13.	Describe indexing and stop-word elimination in Information Retrieval with suitable examples.	CO4	L2	10 M				
14.	Describe stemming in an Information Retrieval system with a suitable example.	CO4	L2	10 M				
15.	Illustrate how the Zipf’s law is used to find the relationship between the frequency of words and their rank order.	CO4	L3	5 M				
16.	Illustrate with example Boolean Information Retrieval model.	CO4	L3	10 M				
17.	Illustrate with example vector space Information Retrieval model.	CO4	L3	10 M				
18.	Consider a document represented by the three terms {tornado, swirl, wind} with the raw tf 4, 1, and 2 respectively. In a collection of 100 documents, 15 documents contain the term tornado, 20 documents contain the term swirl, and 40 documents contain the term wind. Compute the inverse document frequency (idf) and term weight of each term.	CO4	L3	5 M				
19.	Illustrate term weighting and inverse document frequency (IDF) with a suitable example.	CO4	L3	10 M				
20.	Apply simple automatic method for obtaining term frequency, document frequency, Idf and weight for the following documents. <table><tr><td>Document</td><td>Text</td></tr><tr><td></td><td></td></tr></table>	Document	Text			CO4	L3	10 M
Document	Text							

	D1	Data mining techniques and applications.			
	D2	Data mining and machine learning.			
	D3	Applications of data science.			
21.	Describe and illustrate Jaccard's coefficient and cosine coefficient for measuring similarity in IR for the following two binary vectors. x = (1, 0, 0, 0, 0, 0, 0, 0, 0, 0) y = (0, 0, 0, 0, 0, 0, 1, 0, 0, 1)		CO4	L3	10 M
22.	Write a note on non-classical models of IR.		CO4	L2	5 M

Module 5

Q.No.	Questions	CO's	RBT Level	Marks
Unit 1: Lexical Resources				
1.	Explain the main types of lexical resources used in NLP with suitable examples.	CO5	L2	10 M
2.	Illustrate the simplified organization of different senses of the word 'read' in WordNet.	CO5	L3	10 M
3.	Explain the major applications of WordNet in NLP with suitable examples.	CO5	L2	10 M
4.	Illustrate FrameNet concepts using the Communication Frame and its Frame Elements with an example.	CO5	L3	10 M
5.	Explain the major applications of FrameNet in NLP with suitable examples.	CO5	L2	10 M
6.	Explain the characteristics and different types of stemmers with suitable examples.	CO5	L2	10 M
7.	Describe any five part-of-speech taggers.	CO5	L2	10 M
8.	Explain stemmers for Indian languages with suitable examples.	CO5	L2	10 M
9.	Explain stemmers for European languages with suitable examples.			
10.	Summarize the applications of stemming.	CO5	L2	10 M
Unit 1: LLM				
11.	Explain the historical development of NLP that led to the emergence of Large Language Models (LLMs).	CO5	L2	10 M
12.	Describe the advantages and disadvantages of Large Language Models (LLMs) with suitable examples.	CO5	L2	10 M
13.	Describe the major applications of Large Language Models (LLMs) with suitable examples.	CO5	L2	10 M
14.	Explain the sustainability challenges of Large Language Models (LLMs) with suitable examples.	CO5	L2	10 M
15.	Explain Conversational Large Language Models (LLMs), their key features, and applications with suitable examples.	CO5	L2	10 M
16.	Describe examples of Conversational LLMs and their key features.	CO5	L2	10 M